USN. | | | | | | | | | | | [22MTCSE301]

THIRD SEMESTER M.Tech EXAMINATION, DECEMBER - 2023

INFERENTIAL STATISTICS USING R

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.

ii. Missing data may be suitably assumed.

1. a. Differentiate **10**

i. Primary and secondary data

ii. Qualitative and quantitative data

with suitable examples.

b. Represent the following distribution by histogram, locate the mode graphically and also verify the modal value. **10**

0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
4	8	4	6	9	6	3	2	1

2. a. A random variable X has the following probability distribution find k, E(X), $E(X^2)$, Var(X), P(X>1), P(X<1) and P(-2<X<2). **10**

X	-2	-1	0	1	2	3
P(x)	0.1	0.1	0.2	2k	0.3	0.1

b. **I.** Suppose that the number of defective screws produced by a sophisticated machine per day has a Poisson distribution with parameter 2. What is the probability that out of the total produce of a day there is **10**

i. No defective screw

ii. Exactly 2 defective screws

iii. At least one defective screw.

II. 20 wrist watches in a box of 100 are defective. If 10 watches are selected at random. Find the probability that

i. 10 are defective

ii. 10 are good

iii. at most 3 are defective.

3. a. Test the hypothesis that, the average content of containers of a particular lubricant is 10 litres. If the content of the random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3 and 9.8, use 1% level of significance. Find the 99% confidence interval for the mean of the population from which the sample is drawn. ($t_{0.005}$ = 3.250) **10**

b. Two random samples were drawn from two normal populations and their values are: **10**

A: 16, 17, 25, 26, 32, 34, 38, 40, 42

B: 14, 16, 24, 28, 32, 35, 37, 42, 43, 45, 47

Test whether the variance of A is lesser than B at 5% level of significance.

$[F_{0.05}(10,8) = 3.35]$

OR

4. a. Write the differences between Parametric and Non- Parametric tests. **08**

b. Fit a Poisson distribution to the given data and test the goodness of fit at 5% level of significance. ($\chi^2_{0.05,2} = 5.99$) **12**

No. of mistakes in page	0	1	2	3	4	5	6
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5. a. A manufacture uses two methods of production with the following results: **10**

Method I: 51 27 42 27 41 29 27 23 25 35 21 37 37

Method II: 36 19 31 19 25 16 39 25 21 17

Carry out run test to find out whether the two samples are from the same population at 5% level of significance. ($Z_{0.025} = 1.96$).

b. A drug is given to 11 patients and difference in their BP were recorded to be: **10**

Before drug A	112	113	118	120	119	113	110	122	126	115	119
After drug A	116	120	117	125	126	111	111	117	126	112	129

Use the Wilcoxon signed rank test to test the hypothesis that drug has no effect on change of BP at 5% level of significance. ($Z_{0.025} = 1.96$)

OR

6. a. Test the hypothesis that the median HDL cholesterol levels in adult population city A and city B are same. Using the following observation and the Mann-Whitney test at 5% level of significance by: **10**

City A 42 20 51 39 57 60 23

City B 30 42 25 29 35

Given that, the critical value of U for at 5% level of significance for two tailed is 5.

b. Ordinal data is displayed in the table below. Using Friedman test, check whether there is any difference among weeks 1, 2, and 3 at $\alpha = 0.05$. ($\chi^2_{0.05,2} = 5.99$). **10**

Week 1	27	2	4	18	7	9
Week 2	20	8	14	36	21	22
Week 3	34	31	3	23	30	6

7. **a.** To assess the significance of possible variation in performance in a certain test between the convent schools of a city, a common test was given to a number of students taken at random from the fifth class of the schools concerned the results given below: **16**

A	B	C
9	13	14
11	12	13
13	40	17
9	15	7
8	2	9

Discuss the analysis of variance for the given data. ($F_{0.05}$ = 3.89)

b. Calculate the SS $_{total}$ for the following RBD data: **04**

Treatment		
Medication	Exercise	Diet
I1	7	12
10	8	6
8	4	10
14	2	8
13	3	5

OR

8. **a.** A study was conducted to determine the factor that reduces blood pressure the most, medication, diet or exercise. 15 patients at a hospital with comparable levels of high blood pressure are randomly assigned to each treatment group. After eight weeks, the drop in systolic blood pressure for each patient was measured. Use the data below and 5% significance level to construct an ANOVA table to test the claim that all three of the treatments produce the same drop in blood pressure. **10**
($F_{0.05\,(2,12)}$ =3.89).

Treatment		
Medication	Exercise	Diet
11	7	12
10	8	6
8	4	10
14	2	8
10	3	5

b. Analyze the variance in the following Latin square of yields (in kgs) of paddy where A, B, C, D denote the different methods of cultivation. ($F_{0.05}$ (3,6) = 4.76) **10**

D 122	A 121	C 123	B 122
B 124	C 123	A 122	D 125
A 120	B 119	D 120	C 121
C 122	D 123	B 121	A 122

Examine whether the different methods of cultivation have given significantly different yields.